F1 RACING ANALYTICS

SYSTEM ARCHITECTURE AND COMPONENT DESIGN

Project	F1 Racing Analytics — Race Data, Visualized
Source basis	Supplied f1_racing_analytics.html and its embedded Tableau references
Document status	Completed project-specific documentation based on available source evidence

Document basis and evidence boundary

This document is populated from the supplied F1 Racing Analytics index.html. It uses the project's own wording and published KPI values. Where the HTML does not expose underlying data, formulas, workbook details or test results, that limitation is stated explicitly instead of inventing project facts.

Architecture summary

The evidenced architecture has three principal parts: a static website, Tableau Public visualizations and external browser assets. The website presents project information and embeds Tableau views; Tableau Public hosts the analytical dashboard and story.

Component 1 — Website presentation layer

A single HTML page contains the navigation bar, hero section, KPI overview, dashboard section, story section, About section and footer. CSS custom properties define the theme. Bootstrap classes provide responsive layout utilities.

Component 2 — Dashboard visualization

The dashboard is embedded using a Tableau placeholder and an object element. The shared view path is 9Z3NGD45Z. The page describes nine views and identifies reliability, driver/team performance, championship analysis, tyre compounds, country participation and podium analysis.

Component 3 — Story visualization

The story is embedded from Tableau shared view JJ57Z8NDB. Its title is Driver Points Leaderboard. The page describes a Top N parameter that changes how many drivers appear.

Component 4 — Embed initialization

JavaScript iterates over two embed configurations, locates each placeholder, sets responsive dimensions, makes the Tableau object visible and loads https://public.tableau.com/javascripts/api/viz_v1.js asynchronously.

Architecture flow

Visitor browser → F1 Racing Analytics HTML/CSS/JavaScript → Tableau embed API → Tableau Public dashboard/story. Separately, browser loads Bootstrap CSS/JS, Bootstrap Icons and Google Fonts.

Architecture boundary

The supplied HTML does not show a backend server, application API, database, authentication layer or custom data pipeline. Those components are not included in the documented architecture.

Source reference: f1_racing_analytics.html supplied for this documentation task.